

AcciSense: Automated CCTV Crash Detection & Multi-Agency Dispatch System

Complete Presentation Speech, Technical Documentation, Mathematical Models & Future Roadmap

1. EXECUTIVE SUMMARY & 60-SECOND PITCH SPEECH

Presenter Speech:

"Good morning evaluators and members of the panel. Every 24 seconds, a life is lost in a road traffic collision worldwide. The single most critical factor determining survival is emergency response time — known as the **Golden Hour**. Current emergency response systems suffer from manual reporting delays, vague location descriptions, and zero real-time situational awareness.

We created **AcciSense** — an autonomous AI platform that converts standard municipal CCTV cameras, RTSP traffic streams, and mobile video into an intelligent emergency dispatch shield. Powered by a multi-scale YOLOv8 deep learning pipeline, AcciSense detects crashes within 400ms, measures structural damage severity, acquires live device GPS coordinates, and instantly broadcasts emergency payloads directly to dedicated control portals for **Hospitals, Police, and Ambulance EMS teams**. AcciSense doesn't just record accidents — it saves lives automatically."

2. MATHEMATICAL FORMULATIONS & AI DETECTION PIPELINE

A. Intersection over Union (IoU) Collision Metric:

For two vehicle bounding boxes A and B:

$$IoU(A, B) = \text{Area}(A \cap B) / \text{Area}(A \cup B)$$

If $IoU(A, B) \geq 0.08$, an active spatial collision overlap is flagged.

B. Vehicle Structural Damage Surface Score (S_D):

Evaluated inside detected vehicle Region of Interest (ROI) using Canny edge density (δ_E) and intensity variance (σ_I^2):

$$S_D = 0.40 \cdot I(\delta_E > 0.12) + 0.35 \cdot I(\sigma_I^2 > 550) + 0.25 \cdot I(\text{Contours} \geq 2)$$

C. Multi-Scale Central Zoom Detection:

Executes a full-frame scan (conf=0.10) alongside a central 80% zoom crop scan (conf=0.08) to detect small or screen-captured vehicles reliably.

3. SYSTEM ARCHITECTURE SPECS

Component	Technology	Functionality
Deep Learning	PyTorch / YOLOv8	Sensitive vehicle detection & classification
Computer Vision	OpenCV / NumPy	Edge density, multi-scale crop zoom, bounding boxes
Backend Framework	Python FastAPI	Async REST API, database ORM, static hosting
Database	SQLite / SQLAlchemy	Incident log persistence & evidence mapping
Real-Time Engine	WebSockets (ws://)	Zero-latency event broadcast to department portals
Department Portals	HTML5 / Leaflet.js	Dedicated control stations for Hospital, Police, EMS

4. MULTI-AGENCY EMERGENCY MESSAGE DISPATCH PAYLOAD

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===== EMERGENCY ALERT =====
■ ACCIDENT CONFIRMED
■ Time (UTC): 2026-08-11 20:22:21
■ Location: Latitude 17.497985, Longitude 78.1467645
■ Hospital Services: NOTIFIED
■ Police Department: NOTIFIED
■ Ambulance Services: NOTIFIED
=====
```

5. STEP-BY-STEP PRESENTATION SCRIPT FOR VIVA & DEMOS

Slide 1: Problem & Motivation

"Bystanders often take 10 to 15 minutes to report accidents, and manual location reports are frequently inaccurate. AcciSense makes traffic infrastructure self-reporting."

Slide 2: Dual-Stage Computer Vision

"AcciSense doesn't just check distance between cars. It combines bounding box spatial overlap (IoU) with surface damage texture analysis, preventing false alarms in heavy traffic."

Slide 3: Multi-Department Dispatch

"When a crash is confirmed, AcciSense sends direct tailored alerts to three separate web stations: Hospital ER (/hospital), Police Control (/police), and Ambulance EMS (/ambulance)."

6. VIVA & JUDGES' Q&A; DEFENSE GUIDE

Q1: How do you prevent false positives in heavy traffic congestion?

Answer: Normal bumper-to-bumper traffic has spatial overlap but low surface damage scores (smooth metallic reflection). AcciSense requires high structural edge density alongside IoU overlap to confirm a crash.

Q2: How does laptop/device location tracking work?

Answer: AcciSense integrates HTML5 navigator.geolocation and Expo Location services to pull high-accuracy hardware GPS coordinates directly from the laptop/device.

7. FUTURE ENHANCEMENTS & EXPANSION ROADMAP

- 1. Autonomous Drone Swarm Dispatch:** Automatically launches drones from junction stations to hover over crash scenes, streaming 4K aerial video.
- 2. Automatic License Plate Recognition (ALPR):** Reads license plates via OCR to retrieve driver blood type and medical history (EHR) before hospital arrival.
- 3. Smart Green Wave Signal Priority:** Interfaces with SCATS traffic lights to force signals green along the ambulance route.
- 4. 5G C-V2X Highway Warning Broadcast:** Transmits decelerate warnings to nearby connected vehicles within 1km to prevent pileups.
- 5. Edge AI TensorRT Acceleration:** Deploys TensorRT FP16 models directly onto NVIDIA Jetson hardware for 120+ FPS local edge processing.

8. REAL-WORLD DEPLOYMENT & NON-CCTV COVERAGE SOLUTIONS

A. Camera Mounting Geometry: Mounted 6m to 10m high at traffic poles angled downward 30° to 45° with 110° wide-angle FOV covering a 150m detection radius.

B. Non-CCTV Regions & Blind Spots:

- 1. Mobile App & G-Sensors:** Smartphone accelerometers detect impact (>3.5G), capturing crash frames and GPS coordinates automatically.
- 2. Patrol Dashcams:** Police and ambulance vehicles stream live dashcam feeds while patrolling remote highway stretches.

C. Edge Architecture & Bandwidth Optimization: Processing is executed on-edge inside camera housings (NVIDIA Jetson). Video is NEVER continuously uploaded to the cloud — only 2KB JSON alert payloads and single annotated crash evidence images are transmitted.